Practical-1

Consider the experiment to toss a coin three times and count the number of heads. Which of the following sample spaces can be used to describe this experiment?
(a) S = {H, T}
(b) S = {HHH, TTT}
(c) S = {0, 1, 2, 3}
(d) S = {1, 2, 3}
(e) S = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}

Solution:

The experiment is:

Toss a coin three times and count the number of heads.

A **sample space** is the set of all possible outcomes of an experiment. The choice of sample space depends on **how we want to describe the outcomes**.

Let us examine each option one by one.

(a) $\boldsymbol{S = \{H, T\}}$

This sample space contains only two outcomes:

- H= Head
- T= Tail

This is suitable **only for a single coin toss**, not for three tosses.

For three tosses, we need more information than just Hor T.

Therefore,

(a) is NOT a valid sample space.

(b) $\boldsymbol{S = \{}$**HHH**, **TTT**$\boldsymbol{\}}$

This contains only two possible outcomes:

- HHH
- TTT

However, when a coin is tossed three times, there are many other possibilities such as

- HHT
- HTH
- THH
- HTT
- THT

- TTH

Since these outcomes are missing, the sample space is incomplete.

Therefore,

(b) is NOT a valid sample space.

(c) S = {0, 1, 2, 3}

The experiment asks us to **count the number of heads**.

Possible numbers of heads are

- 0 heads (TTT)
- 1 head (HTT, THT, TTH)
- 2 heads (HHT, HTH, THH)
- 3 heads (HHH)

Thus every possible result is represented.

Outcome	**Number of Heads**
TTT	0
HTT, THT, TTH	1
HHT, HTH, THH	2
HHH	3

Hence,

$$S = \{0,1,2,3\}$$

is a perfectly valid sample space **because the experiment records only the number of heads.**

Therefore,

(c) is a valid sample space.

(d) S = {1, 2, 3}

This sample space omits the outcome

- **0 heads**, which occurs when all tosses are tails (TTT).

Since one possible outcome is missing, the sample space is incomplete.

Therefore,

(d) is NOT a valid sample space.

(e) $\mathbf{S} = \{\mathbf{HHH}, \mathbf{HHT}, \mathbf{HTH}, \mathbf{THH}, \mathbf{HTT}, \mathbf{THT}, \mathbf{TTH}, \mathbf{TTT}\}$ This lists **every possible sequence** of three-coin tosses. Since each toss has two outcomes,

$$2^3 = 8$$

possible sequences exist.

Outcome	**Heads**
HHH	3
HHT	2
HTH	2
THH	2
HTT	1
THT	1
TTH	1
TTT	0

Every possible outcome is included exactly once.

Hence,

(e) is also a valid sample space.

Although the experiment asks us to count the number of heads, using the complete list of sequences is still a correct sample space because the number of heads can be determined from each sequence.

Final Answer

The valid sample spaces are

$$(c)\ S = \{0,1,2,3\}$$

and

(*e*) S = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}

Summary Table

Option	**Valid?**	**Reason**
(a) ({H,T})	No	Suitable only for one toss.
(b) ({HHH,TTT})	No	Many outcomes are missing.
(c) ({0,1,2,3})	Yes	Represents all possible numbers of heads.
(d) ({1,2,3})	No	Omits the outcome of 0 heads.
(e) ({HHH,HHT,HTH,THH,HTT,THT,TTH,TTT})	Yes	Contains all 8 possible outcomes of three tosses.

Answer:

The correct sample spaces are (c) and (e).

Practical-2

Find out the followings:
(a) If you roll two fair six-sided dice, what is the probability that the sum of the number rolled is 7?
(b) A card is drawn from a standard deck. What is the probability it's a king given it's a face card?

Solution:

The probability of an event is given by

$$\text{Probability} = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

We solve each part separately.

(a) If you roll two fair six-sided dice, what is the probability that the sum of the numbers rolled is 7?

Step 1: Find the sample space

A fair die has six faces:

$$\{1,2,3,4,5,6\}$$

When two dice are rolled, each die can show any one of the six numbers.

Hence, the total number of possible outcomes is

$$6 \times 6 = 36.$$

The sample space consists of the following 36 ordered pairs:

$$\begin{aligned} S=\{&(1,1),(1,2),(1,3),(1,4),(1,5),(1,6),\\ &(2,1),(2,2),(2,3),(2,4),(2,5),(2,6),\\ &(3,1),(3,2),(3,3),(3,4),(3,5),(3,6),\\ &(4,1),(4,2),(4,3),(4,4),(4,5),(4,6),\\ &(5,1),(5,2),(5,3),(5,4),(5,5),(5,6),\\ &(6,1),(6,2),(6,3),(6,4),(6,5),(6,6)\}. \end{aligned}$$

Therefore, $n(S) = 36$.

Step 2: Find the favorable outcomes

We need the outcomes whose **sum is 7**.

These are:

Outcome	**Sum**
(1,6)	7
(2,5)	7
(3,4)	7
(4,3)	7
(5,2)	7
(6,1)	7

Thus,

$$n(E) = 6.$$

Step 3: Apply the probability formula

$$P(E) = \frac{n(E)}{n(S)} = \frac{6}{36} = \frac{1}{6}.$$

Final Answer (a)

$$\boxed{P(\text{sum} = 7) = \frac{1}{6}}$$

(b) A card is drawn from a standard deck. What is the probability it is a king given it is a face card?

This is a **conditional probability** problem.

Step 1: Recall the standard deck

A standard deck has

- 52 cards
- 4 suits:
    - Hearts
    - Diamonds
    - Clubs
    - Spades

- Each

suit contains

- 1 Jack
- 1 Queen
- 1 King

These are called **face cards**.

Hence,

Number of face cards

$$4 + 4 + 4 = 12.$$

Step 2: Number of kings

Number of kings

$$= 4.$$

Since every king is a face card,

$$\text{Kings among face cards} = 4.$$

Step 3: Use conditional probability

The formula is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

where

- A: Card is a King
- B: Card is a Face card

Since every king is a face card,

$$P(\text{King} \mid \text{Face card}) = \frac{4}{12} = \frac{1}{3}.$$

Alternative Method

Restrict the sample space to only face cards.

Face cards are

- 4 Jacks
- 4 Queens
- 4 Kings

Total face cards

12.

Among these,

Kings 4.

Therefore,

$$P(\text{King} \mid \text{Face card}) = \frac{4}{12} = \frac{1}{3}.$$

Final Answer (b)

$$\boxed{P(King \mid Face\ card) = \frac{1}{3}}$$

Practical-3:

Let A, B, and C be three events. Express the following events in terms of A, B, and C: (a) None of the events occurs. (b) At least one of the events occurs. (c) All of the events occur.

Solution

Let **A, B, and C** be three events in a sample space S.

We are required to express each statement using the operations of sets (events):

- **Union (∪)** → "or" (at least one event occurs)
- **Intersection (∩)** → "and" (all events occur)
- **Complement (^{c} or $'$)** → "does not occur"

(a) None of the events occurs

Step 1: Understand the

statement

The statement **"None of the events occurs"** means:

- Event Adoes **not** occur.
- Event Bdoes **not** occur.
- Event Cdoes **not** occur.

These three conditions must happen simultaneously.

Step 2: Write in mathematical form

The complement of an event means the event does not occur.

- A^c= A does not occur
- B^c= B does not occur
- C^c= C does not occur

Since all three must occur together,

$$\boxed{A^c \cap B^c \cap C^c}$$

Step 3: Alternative expression

Using **De Morgan's Law**,

$$(A \cup B \cup C)^c = A^c \cap B^c \cap C^c.$$

Thus,

$$\boxed{(A \cup B \cup C)^c}$$

is an equivalent expression.

Final Answer (a)

$$\boxed{A^c \cap B^c \cap C^c}$$

or equivalently,

$$\boxed{(A \cup B \cup C)^c.}$$

(b) At least one of the events occurs Step 1: Understand the statement "At least one" means

- Only A occurs, or
- Only B occurs, or
- Only C occurs, or
- Any two occur, or
- All three occur.

In short, **one or more** of the events occurs.

Step 2: Write mathematically

The union of events represents "at least one occurs."

Therefore,

$$\boxed{A \cup B \cup C}$$

Alternative expression

Using complements,

$$A \cup B \cup C = (A^c \cap B^c \cap C^c)^c.$$

Final Answer (b)

$$\boxed{A \cup B \cup C}$$

(c) All of the events occur

Step 1: Understand the statement

The statement means

- A occurs,
- B occurs,
- C occurs,

all at the same time.

Step 2: Write mathematically

The intersection of events means all occur simultaneously.

Therefore,

$$\boxed{A \cap B \cap C}$$

Summary Table

Statement	**Mathematical Expression**	**Explanation**
(a) None of the events occurs	$A^c \cap B^c \cap C^c$	A, B, and C all do not occur.
(b) At least one of the events occurs	$A \cup B \cup C$	One or more of A, B, or C occurs.
(c) All of the events occur	$A \cap B \cap C$	A, B, and C occur together.

Practical-4:

Write a code to simulate coin tosses and calculate the probabilities of getting heads and tails based on the number of tosses are specified by the user.

Solution:
Program to Simulate Coin Tosses and Calculate the Probabilities of Heads and Tails

Aim

To write a program that simulates the tossing of a fair coin $n$times (where $n$is specified by the user) and calculates the experimental probabilities of obtaining **Heads** and **Tails**.

Theory

A **coin toss** is a random experiment in which there are only two possible outcomes:

- **Head (H)**
- **Tail (T)**

If the coin is fair, then theoretically,

$$P(\text{Head}) = \frac{1}{2}$$

and

$$P(\text{Tail}) = \frac{1}{2}$$

However, when the experiment is performed a finite number of times, the observed probabilities may differ slightly from the theoretical probability.

The experimental probability is calculated as

$$P(\text{Head}) = \frac{\text{Number of Heads Total Tosses}}{}$$

and

$$P(\text{Tail}) = \text{N u m b e r o f T a}$$

Algorithm Step 1: Start.

- Head = Head + 1

- Otherwise,
  - Tail = Tail + 1

Step 5: Calculate $P(H) = \frac{\text{Heads}}{n}$

$P(T) = \frac{\text{Tails}}{n}$

Step 6: Display

- Number of Heads
- Number of Tails
- Probability of Head
- Probability of Tail

Step 7: Stop.

MATLAB Program

```
clc;
clear;

% Input number of tosses
n = input('Enter the number of coin tosses: ');

% Initialize counters
heads = 0;
tails = 0;

% Simulate coin tosses
for i = 1:n

    toss = randi([0 1]);      % 0 = Head, 1 = Tail
```

```
    if toss = 0
        heads = heads + 1;
    else
        tails = tails + 1;
    end

end

% Calculate
probabilities P_head =
heads / n; P_tail = tails /
n;

% Display results
fprintf('\nNumber of Heads = %d\n', heads);
fprintf('Number of Tails = %d\n', tails);

fprintf('\nExperimental Probability of Head = %.4f\n', P_head);
fprintf('Experimental Probability of Tail = %.4f\n', P_tail);
```

```
Enter the number of coin tosses: 10

Number of Heads = 6
Number of Tails = 4

Experimental Probability of Head = 0.6000
Experimental Probability of Tail = 0.4000
```

Explanation of the Program

n = input('Enter the number of coin tosses: ');
Reads the total number of tosses from the user.
heads = 0;
tails = 0;

Initializes the counters for heads and tails.
for i = 1:n
Repeats the experiment **n** times. toss = randi([0 1]);
Generates a random integer.

- 0 → Head
- 1 → Tail

Each outcome has an equal probability of occurring.

if toss == 0

Checks whether the toss resulted in a Head.

If true,

heads = heads + 1;

Otherwise,

tails = tails + 1;

After completing all

tosses, After completing

all tosses, P_head =

heads/n;

calculates the experimental probability of obtaining a Head.

Similarly,

P_tail = tails/n;

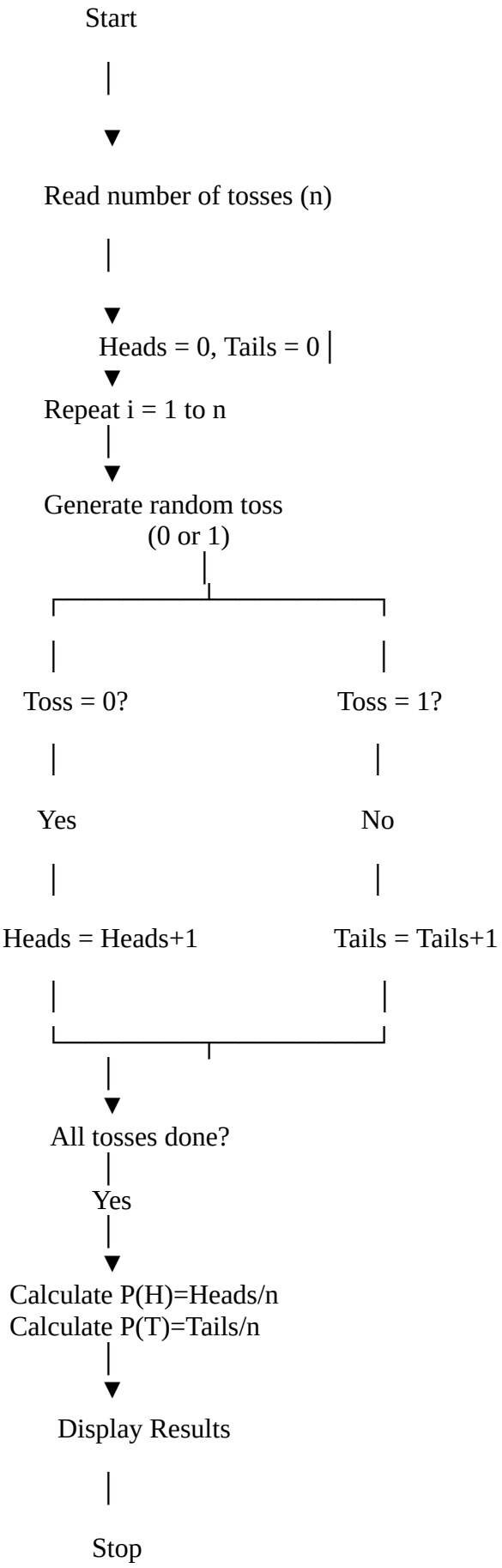
calculates the experimental probability of obtaining a Tail.

Finally,

fprintf(...)

prints the results on the screen.

Flowchart


Start
Read number of tosses (n)
Heads = 0, Tails = 0
Repeat i = 1 to n
Generate random toss
(0 or 1)
Toss = 0?
Toss = 1?
Yes
No
Heads = Heads+1
Tails = Tails+1
All tosses done?
Yes
Calculate P(H)=Heads/n
Calculate P(T)=Tails/n
Display Results
Stop

Practical 5

In a city, 60% of the population owns a car, and 40% of car owners are under 30 years old. Additionally, 30% of the population is under 30 years old. If a randomly selected person from the city is under 30 years old, what is the probability that this person owns a car?

Solution:

This problem is based on **Bayes' Theorem**, which is used to find the probability of an event based on prior information.

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)} = \frac{0.85(0.20)}{0.25} = 0.68$$

Where P(A) = 0.20, P(B|A) = 0.85, P(A) = 0.25

Step 1: Define the Events

Let

- A: The person **owns a car**.
- B: The person is **under 30 years old**.

We are given:

- Probability that a person owns a car:

$$P(A) = 60\% = 0.60$$

- Probability that a car owner is under 30:

$$P(B \mid A) = 40\% = 0.40$$

- Probability that a randomly selected person is under 30:

$$P(B) = 30\% = 0.30$$

We are asked to find

$$P(A \mid B),$$

which is the probability that a person **owns a car given that the person is under 30 years old**.

Step 2: Write Bayes' Theorem

Bayes' Theorem states:

$$P(A \mid B) = \frac{P(B \mid A) \times P(A)}{P(B)}$$

Step 3: Substitute the Given Values

Substitute the given probabilities into the formula:

$$P(A \mid B) = \frac{0.40 \times 0.60}{0.30}$$

Step 4: Perform the Calculations

First, multiply the numerator:

$$0.40 \times 0.60 = 0.24$$

Now divide by 0.30:

$$P(A \mid B) = \frac{0.24}{0.30} = 0.80$$

Step 5: Convert to Percentage

$$0.80 \times 100 = 80\%$$

Verification Using Population Numbers

Assume the city has **100 people**.

- Car owners:

$$60\% \text{ of } 100 = 60$$

- Under-30 car owners:

$$40\% \text{ of } 60 = 24$$

- Total people under 30:

$$30\% \text{ of } 100 = 30$$ Among the **30 people under 30**, **24 own a car**. Therefore,

$$P(\text{Car Owner} \mid \text{Under 30}) = = 0.80.$$

This confirms the answer.

Interpretation

Although only **60%** of the city's population owns a car, among people **under 30 years old**, the probability of owning a car is **80%**. This is because a relatively large proportion of car owners are under 30.

Final Answer

$$\boxed{P(\text{Car Owner} \mid \text{Under 30}) = 0.80}$$

or

$$\boxed{80\%}$$

Summary Table

Quantity	**Value**
P (Car Owner)	(0.60)
P (Under 30|Car Owner)	(0.40)
P (Under 30)	(0.30)
P(Car Owner | Under 30)	0.80(80%)

Practical 6

Write a code to simulate rolling a fair die and calculate the probabilities of each face appearing where number of rolls is specified by the user.

Solution:

Program to Simulate Rolling a Fair Die and Calculate the Probabilities of Each Face

Aim

To write a program that simulates rolling a **fair six-sided die** for a number of times specified by the user and calculates the **experimental probability** of each face (1 to 6).

Theory

A **fair die** has six faces numbered

$$\{1, 2, 3, 4, 5, 6\}$$

Each face has an equal chance of occurring.

The **theoretical probability** of obtaining any face is

$$P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}$$

or

$$\boxed{\frac{1}{6} \approx 0.1667}$$

When the die is rolled many times, the **experimental probability** of each face is

$$\boxed{P(i) = \frac{\text{Number of times face } i \text{ appears}}{\text{Total number of rolls}}}$$

where

$$i = 1,2,3,4,5,6.$$

As the number of rolls increases, the experimental probabilities approach the theoretical probability $\frac{1}{6}$.

Algorithm

Step 1: Start.

Step 2: Read the number of die rolls n.

Step 3: Initialize counters for faces 1 to 6.

Step 4: Repeat from 1 to n:

- Generate a random integer between 1 and 6.
- Increase the corresponding face counter by 1.

Step 5: Calculate the experimental probability of each face:

$$P(i) = \frac{\text{Count of face } i}{n}$$

Step 6: Display

- Frequency of each face.
- Probability of each face.

Step 7: Stop.

MATLAB Program

```
clc;
clear;
% Input number of rolls

n = input('Enter the number of die rolls: ');
% Initialize counters
count = zeros(1,6);
% Simulate die rolls
for i = 1:n
   roll = randi([1 6]);      % Random integer between 1 and 6
   count(roll) = count(roll) + 1;
end
% Calculate probabilities

probability = count / n;

% Display frequencies

fprintf('\nFace\tFrequency\tProbability\n'

or i = 1:6

fprintf('%d\t%d\t\t%.4f\n',i,count(i),probability(i));

End
```

```
Enter the number of coin tosses: 10

Number of Heads = 6
Number of Tails = 4

Experimental Probability of Head = 0.6000
Experimental Probability of Tail = 0.4000
```

Explanation of the MATLAB Code

Step 1: Clear the Workspace

clc;

clear

;

- clc clears the Command Window.
- clear removes all variables from memory.

Step 2: Read Number of Rolls

n = input('Enter the number of die rolls: ');

The user enters the total number of die rolls.

Example:

n = 100

Step 3: Initialize the Counter

count = zeros(1,6);

Creates a vector

count = [0 0 0 0 0 0]

Each element stores the frequency of one face.

Step 4: Simulate Rolling the Die

roll = randi([1 6]);

randi([1 6]) generates a random integer between **1 and 6**.

For example,

3 1 6 2 5

Each number represents the face obtained in one roll.

Step 5: Count the Occurrences

count(roll)=count(roll)+1;

Suppose

roll = 4

then

count(4) = count(4)+1

increments the frequency of face 4.

Step 6: Calculate Experimental Probabilities

probability = count/n;

If

count = [18 16 17 15 19 15]

and

n = 100

then

probability =

0.18

0.16

0.17

0.15

0.19

0.15

Step 7: Display Results

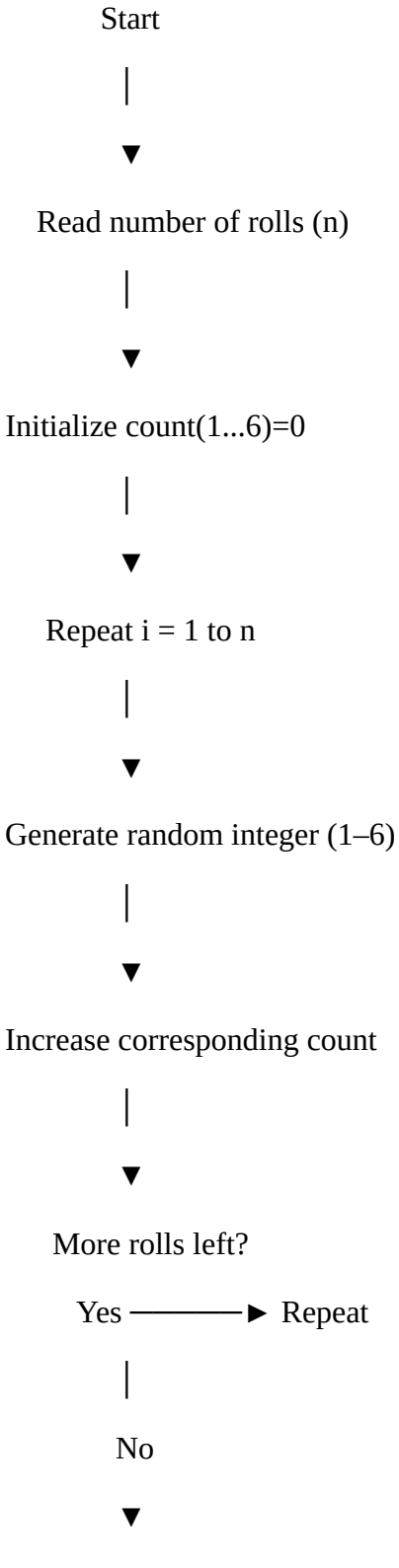
fprintf(...)

Prints a table containing

- Face value

- Frequency
- Experimental probability

Flowchart





Display Frequencies

and Probabilities



Stop

Theoretical vs Experimental Probability

Face	Theoretical Probability	Experimental Probability (Example)
1	$\frac{1}{6} = 0.1667$	0.1690
2	$\frac{1}{6} = 0.1667$	0.1660
3	$\frac{1}{6} = 0.1667$	0.1700
4	$\frac{1}{6} = 0.1667$	0.1610
5	$\frac{1}{6} = 0.1667$	0.1670
6	$\frac{1}{6} = 0.1667$	0.1670

As the number of rolls increases, the experimental probabilities become closer to the theoretical value of $\frac{1}{6}$.

Practical 7

Jamie's dad gave her a die for her birthday. She wanted to make sure it was fair so she took her die to school and rolled it 500 times and kept track of how many times the die rolled each number. Afterwards she calculated the expected value of the sum of 20 rolls to be 67.4.
On her way home from school it was raining and 2 values were washed away from her data table.
Find the 2 missing absolute frequencies from Jamie's data table.

1. Expected Value for Discrete Random Variables

If X is a discrete random variable with possible values $x_1, x_2, \ldots, x_n$ and associated probabilities $P(X = x_i)$, then the expected value is given by:

$$E[X] = \sum_{i=1}^{n} x_i \cdot P(X = x_i)$$

Example

Suppose X is the outcome of rolling a 6-sided die. The possible outcomes are $x = 1, 2, 3, 4, 5, 6$, each with probability $\frac{1}{6}$. The expected value is:

$$E[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6}$$

$$E[X] = \frac{21}{6} = 3.5$$

This means the average outcome over many rolls would be **3.5**.

Jamie's dad gave her a die for her birthday. She wanted to make sure it was fair so she took her die to school and rolled it 500 times and kept track of how many times the die rolled each number. Afterwards she calculated the expected value of the sum of 20 rolls to be 67.4.

On her way home from school it was raining and 2 values were washed away from her data table.

Find the 2 missing absolute frequencies from Jamie's data table.

Die value	Absolute frequency
1	A = 75
2	110
3	95
4	70
5	75
6	B = 75
Total	500

$$500\left(\frac{A}{500}(1) + \frac{110}{500}(2) + \frac{95}{500}(3) + \frac{70}{500}(4) + \frac{75}{500}\cdot 5 + \frac{B}{500}\cdot 6\right) = 3.37(500)$$

$$A + 220 + 285 + 280 + 375 + 6B = 1685$$

$$A + 6B = 525$$

$$-A + -B = -150$$

$$5B = 375$$

$$B = 75$$

Exp value of roll = $\frac{67.4}{20}$

$= 3.37$

$A + 110 + 95 + 70 + 75 + B = 500$

$A + 75 = 150$

$A = 75$

Practical 8

Find out the following:

(a) Consider tossing the coin for 15 times, what is the probability of getting heads for 7 times?

A manufacturing company produces light bulbs, and each light bulb has a probability of 0.4 of being non-defective and a probability of 0.6 of being defective. What is the probability of producing exactly 2 non-defective light bulbs if total produced bulbs are 5?

Solution

Both parts of this question are based on the **Binomial Distribution**.

A binomial distribution is used when:

1. There is a fixed number of trials (n).
2. Each trial has only two possible outcomes (success or failure).
3. The probability of success (p) remains constant.
4. The trials are independent.

The probability of obtaining exactly $x$successes in $n$trials is given by the **Binomial Probability Formula**:

$$P(X = x) = \binom{n}{x} p^{x}(1-p)^{n-x}$$

where

- n= Number of trials
- x= Number of successes
- p= Probability of success
- $1 - p = q$= Probability of failure

(a) Consider tossing a coin 15 times. What is the probability of getting exactly 7 heads? Step 1: Identify the given data

Number of tosses

$$n = 15$$

Number of heads required

$$x = 5 = 1 - p = 0.5$$

Step 2: Apply the Binomial Formula

$$P(X = 7) = \binom{15}{} (0.5)^{7}(0.5)^{8}$$

Since

$$(0.5)^7(0.5)^8 = (0.5)^{15},$$

we get

$$P(X = 7) = \binom{15}{7} (0.5)^{15}.$$

Step 3: Calculate the Combination

$$\binom{15}{7} = \frac{15!}{7!\,8!} = 6435.$$

Also,

$$(0.5)^{15} = \frac{1}{32768} = 0.000030517\overline{6}.$$

Step 4: Calculate the Probability

$$P(X = 7) = 6435 \times 0.0000305176 = 0.19638.$$

Final Answer (a)

$$\boxed{P(\text{Exactly 7 heads}) \approx 0.1964}$$

or

$$\boxed{19.64\%}$$

(b) A manufacturing company produces light bulbs. Each bulb has a probability of 0.4 of being non-defective and 0.6 of being defective. If 5 bulbs are produced, what is the probability of getting exactly 2 non-defective bulbs?

Step 1: Identify the given data

Number of bulbs

$$n = 5$$

Required non-defective bulbs

$$x = 2$$

Probability of non-defective bulb

$$p = 0.4$$

Probability of defective bulb

$$q = 0.6$$

Step 2: Apply the Binomial Formula

$$P(X = 2) = \binom{5}{2} (0.4)^2(0.6)^3.$$

Step 3: Calculate Each

Quantity Combination

$$\binom{5}{2} = \frac{5!}{2!\,3!} = 10.$$

Probability terms

$$(0.4)^2 = 0.16$$
$$(0.6)^3 = 0.216.$$

Step 4: Calculate the Probability

$$P(X = 2) = 10(0.16)(0.216)$$
$$= 10(0.03456)$$
$$= 0.3456.$$

Final Answer (b)

$$\boxed{P(\text{Exactly 2 non-defective bulbs}) = 0.3456}$$

or

$$\boxed{34.56\%}$$

Summary Table

Part	**Number of Trials ((n))**	**Successes ((x))**	**Probability of Success ((p))**	**Probability**
(a) Coin tossed 15 times, exactly 7 heads	15	7	0.5	0.1964
(b) 5 bulbs, exactly 2 non-defective	5	2	0.4	0.3456

Final Answers

(a)

$$\boxed{P(\text{Exactly 7 heads in 15 tosses}) \approx 0.1964 \ (19.64\%).}$$

(b)

$$\boxed{P(\text{Exactly 2 non-defective bulbs out of 5}) = 0.3456 \ (34.56\%).}$$

Both problems use the **Binomial Distribution**, where the probability of obtaining exactly $x$successes in $n$independent trials is calculated using

$$\boxed{P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}.}$$

Practical 9

Assume that, scores of students in math exam follow a normal distribution with mean marks 70 and standard deviation 10. What is the probability that randomly chosen student scores above 85 marks?

Solution:

This problem is based on the **Normal Distribution**.

The marks are normally distributed with:

- Mean:

$$\mu = 70$$

- Standard deviation:

$$\sigma = 10$$

We are required to find the probability that a randomly selected student scores **more than 85 marks**.

That is,

$$P(X > 85).$$

Step 1: Write the Given Data

Mean

$$\mu = 70$$

Standard deviation

$$\sigma = 10$$

Required score

$$X = 85.$$

Step 2: Convert the Score into a Standard Normal Variable

To find the probability, convert the score to a **Z-score** using

$$z = \frac{x - \mu}{\sigma}.$$

Substitute the given values:

$$z = \frac{85 - 70}{10} = \frac{15}{10} = 1.5.$$

Thus, $\boxed{z = 1.5}$

Step 3: Use the Standard Normal Distribution Table

From the standard normal (Z) table,

$$P(Z < 1.5) = 0.9332.$$

This represents the probability that a student's score is **less than 85**.

Step 4: Find the Required Probability

We need

$$P(X > 85) = P(Z > 1.5).$$

Using the complement rule,

$$P(Z > 1.5) = 1 - P(Z < 1.5).$$

Substitute the value from the Z-table:

$$P(Z > 1.5) = 1 - 0.9332 = 0.0668.$$

Step 5: Convert to Percentage

$$0.0668 \times 100 = 6.68\%.$$

Final Answer

$$\boxed{P(X > 85) = 0.0668}$$

or

$$\boxed{6.68\%}$$

Summary Table

Quantity	**Value**
Mean (μ)	70
Standard deviation (σ)	10
Required score (X)	85
Z-score	1.5
(P(Z<1.5))	0.9332
(P(Z>1.5))	0.0668
Probability in percentage	6.68%

Practical 10

Assume that, scores of students in math exam follow a normal distribution with mean marks 70 and standard deviation 10. What is the probability that randomly chosen student scores between 60 and 80 marks?

Solution

This problem is based on the **Normal Distribution**.

The marks of students are normally distributed with:

- Mean:

$$\mu = 70$$

- Standard deviation:

$$\sigma = 10$$

We are required to find the probability that a randomly selected student scores **between 60 and 80 marks**.

That is,

$$P(60 < X < 80).$$

Step 1: Write the Given Data

Mean

$$\mu = 70$$

Standard deviation

$$\sigma = 10$$

Required interval

$$60 < X < 80$$

Step 2: Convert the Scores into Z-scores

Use the standard normal transformation: $z = \dfrac{x - \mu}{\sigma}$.

μμ−1σμ+1σ

For $X = 60$

$$z = \frac{60 - 70}{10} = \frac{-10}{10} = -1.$$

For $X = 80$

$$z = \frac{80 - 70}{10} = \frac{10}{10} = 1.$$

Thus,

$$\boxed{P(60 < X < 80) = P(-1 < Z < 1).}$$

Step 3: Use the Standard Normal Distribution Table

From the standard normal (Z) table,

$$P(Z < 1) = 0.8413$$

and

$$P(Z < -1) = 0.1587.$$

Step 4: Find the Required Probability

The probability between the two Z-values is

$$P(-1 < Z < 1) = P(Z < 1) - P(Z < -1).$$

Substitute the values:

$$P(-1 < Z < 1) = 0.8413 - 0.1587 = 0.6826.$$

Step 5: Convert to Percentage

$$0.6826 \times 100 = 68.26\%.$$

Final Answer

$$\boxed{P(60 < X < 80) = 0.6826}$$

or

$$\boxed{68.26\%}$$

Summary Table

Quantity	**Value**
Mean (μ)	70

Quantity	**Value**
Standard deviation (σ)	10
Lower score	60
Upper score	80
Lower Z-score	(-1)
Upper Z-score	(1)
(P(Z<1))	0.8413
(P(Z<-1))	0.1587
(P(60<X<80))	0.6826
Probability in percentage	68.26%